Research Paper 1: A Smart Home Energy Management System Using IoT and Big Data Analytics Approach

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Abstract

This paper presents a Smart Home Energy Management System (EMS) that uses the Internet of Things (IoT) and Big Data analytics to monitor, control and improve electricity consumption. The main idea is to connect household appliances to a network so that energy data can be collected continuously. The collected information is sent to a central server for analysis. The system can also provide remote access so that users can monitor appliance usage and switch devices ON or OFF. The paper demonstrates the approach using residential HVAC equipment as a case study.

Introduction

Increasing energy demand and electricity cost make efficient energy management important. Traditional systems mainly measure electricity use, but they do not always provide detailed information about individual appliances or allow intelligent control. The authors propose an IoT-based system in which appliances act as connected devices. This creates a platform for collecting energy data and using it to make better decisions.

Proposed System

Each home appliance is connected to a data-acquisition module. The module measures energy consumption and communicates through a wireless network. Every connected device can be identified through its network address. The collected data is transmitted to a centralized server where it can be stored and analyzed. Big Data and Business Intelligence tools are used to understand consumption patterns and support energy-management decisions. Remote access also allows users to view appliance status and control devices through a computer or mobile device.

Working

The basic working sequence is: appliance → sensing/data-acquisition module → wireless communication → central server → data analysis → user/control action. The system continuously gathers consumption information. After analysis, users can identify high-consumption devices and change their usage. Remote ON/OFF control helps reduce unnecessary operation of appliances.

Case Study and Findings

HVAC equipment was selected as a practical case study because air-conditioning can represent a large portion of electricity consumption in some regions. A prototype was developed and tested in a laboratory environment. The study shows that combining IoT connectivity with centralized data analysis can provide better visibility and control of household energy use.

Advantages

- Real-time or continuous energy data collection.
- Remote monitoring and appliance control.
- Centralized storage and analysis of consumption data.
- Helps identify high-energy-consuming appliances.
- Can support better energy-saving decisions.
- Can be extended to residential, commercial and industrial environments.

Limitations

The system depends on reliable communication between devices and the central server. Large numbers of connected devices also produce large amounts of data that require suitable storage, processing and security. The

approach can require additional hardware and networking infrastructure.

Conclusion

The paper demonstrates that IoT and data analytics can make energy management smarter than simple electricity measurement. By collecting appliance-level information and providing remote control, the proposed system can help users understand their consumption and reduce unnecessary power usage. This concept is directly useful for a Smart Power Management project using sensors, a microcontroller and a web/mobile interface.

Note: This is an original student-friendly summary of the cited research paper, prepared for SPI academic submission; it is not a reproduction of the original paper.

Research Paper 2: Design and Implementation of Smart Home Energy Management System for Indian Residential Sector

Authors/Source: Research article in Energy Conversion and Management

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Abstract

This research presents an IoT-enabled Home Energy Management System designed for the Indian residential sector. The system focuses on monitoring household electricity use and managing loads according to usage patterns. A programmable load-categorization method is introduced to support peak-load management. The study also considers consumer readiness for demand response. The reported results show potential electricity-cost savings of up to 49% under a fixed tariff and 53% under a Time-of-Day tariff.

Introduction

Residential electricity consumption is increasing because homes use more electrical and smart appliances. During peak periods, high demand can increase electricity costs and put additional pressure on the power system. A Home Energy Management System can help by monitoring loads and scheduling flexible appliances. This paper is particularly useful for an Indian project because it considers the residential conditions and tariff environment of India.

Proposed Approach

The system uses IoT technology to monitor and manage household electrical loads. Appliances are categorized according to their characteristics and ability to be shifted or controlled. Loads that do not need to operate at a specific time can be scheduled for more suitable periods. This reduces unnecessary peak demand while maintaining the user's essential requirements.

Load Management

The key idea is programmable load categorization. Essential appliances can continue to operate when required, while flexible appliances can be shifted from high-demand periods to lower-demand periods. Time-of-Day electricity pricing can further encourage users to operate flexible loads when electricity is cheaper. Such scheduling is an important part of smart power management.

Consumer and Practical Considerations

The research also evaluates energy-usage patterns and consumer readiness for demand-response programs. A smart energy system is not useful only because of its technical design; users must be able to understand and accept automatic or scheduled control. Therefore, ease of use, convenience and clear information about energy consumption are important parts of the system.

Results

The study reports cost savings of up to 49% with a fixed tariff and up to 53% with a Time-of-Day tariff. These results indicate that intelligent scheduling and demand-side management can provide meaningful economic benefits. The exact savings depend on appliance usage, tariff structure and user behaviour.

Advantages

- Designed with Indian residential conditions in mind.
- Uses IoT for smart energy management.
- Supports peak-load reduction.
- Allows programmable categorization of loads.
- Can reduce electricity cost through better scheduling.
- Connects technical energy management with user behaviour.

Limitations

The effectiveness depends on correct load information, suitable appliance scheduling and user acceptance. Real-world savings can vary between households because electricity consumption patterns and appliance ownership are different. Internet connectivity and compatible control hardware may also be required.

Conclusion

The paper shows that a smart home energy management system can help Indian households reduce peak demand and electricity cost. Its concept of categorizing appliances and scheduling flexible loads is highly suitable for a student Smart Power Management project. A simplified prototype can use sensors, a microcontroller, relays and a dashboard to demonstrate the same basic idea.

Note: This is an original student-friendly summary of the cited research paper, prepared for SPI academic submission; it is not a reproduction of the original paper.

Research Paper 3: State-of-the-Art and Real-Time Implementation of an IoT-Based Home Energy Management System for a Cluster of Dwellings

Authors/Source: Research article by authors from VIT and collaborators

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Abstract

This paper presents a real-time IoT-based Home Energy Management System for managing electricity consumption across homes. The study focuses on three important goals: reducing electricity cost, reducing the peak-to-average ratio (PAR), and maintaining user convenience. A smart plug and a principal load scheduler are used to implement and test the proposed scheduling approach. The system shifts suitable appliance loads away from peak periods and validates the approach using hardware.

Introduction

Electricity demand varies during the day, and peak demand can increase electricity cost and create stress on the power grid. Home Energy Management Systems can help by scheduling appliances at suitable times. However, an effective system should not only reduce cost; it should also control peak demand without making users uncomfortable. The research therefore considers multiple objectives instead of focusing on only one measurement.

System Architecture

The proposed solution combines IoT communication, smart plugs, a load scheduler and a mobile application. Smart plugs provide control and energy-related information for appliances. The scheduler decides when flexible appliances should operate. The mobile application provides an interface for users. The system can therefore connect energy measurement, automatic scheduling and user interaction.

Load Scheduling

Appliances are considered according to whether their operation can be shifted. Flexible loads can be moved from peak hours to off-peak hours. The paper uses a multi-objective optimization approach called Multi-Objective Mountain Gazelle Optimization (MMGO) to generate suitable schedules. The purpose is to balance electricity cost, peak demand and user convenience.

Experimental Results

The proposed approach was validated using a hardware prototype with a smart plug. For a single dwelling, the reported results show a 12.14% reduction in electricity cost and a 52.54% reduction in peak-to-average ratio compared with unscheduled loads. During peak hours, the reported cost reduction reached up to 80.15%. For a cluster of 50 homes, the study reports overall reductions of about 11% in cost and 74.68% in peak-to-average ratio.

Importance for Smart Power Management

The study is useful because it moves beyond a theoretical model and demonstrates real-time implementation. It shows that smart power management can combine sensors, connected plugs, scheduling algorithms and a mobile interface. For a student prototype, the complex optimization algorithm can be simplified into rule-based scheduling while keeping the same basic concept of shifting non-essential loads.

Advantages

- Real-time IoT-based monitoring and control.
- Uses smart plugs for practical appliance control.
- Considers both cost and peak demand.
- Provides a mobile interface for users.
- Includes hardware-based validation.

• Can be extended from one home to multiple homes.

Limitations

Optimization-based scheduling can be more complex than simple automatic ON/OFF control. The system also depends on connected hardware and accurate consumption information. User preferences and appliance operating requirements must be considered when creating schedules.

Conclusion

The research demonstrates that IoT-based load scheduling can reduce electricity cost and peak demand while keeping user convenience in consideration. It provides a strong reference for Smart Power Management projects because it combines monitoring, automatic control, scheduling and a user interface. A simplified student model can implement these functions using an ESP32/Arduino, current sensor, relay module and mobile or web dashboard.

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